Mathematical Modeling Lecture Notes
Chapter 10 – Game Theory

I. Total conflict games
   a. Introductory example: Rock, Paper, Scissors
      i. In this game, rock beats scissors, scissors beat paper, and paper beats rock.
      ii. So, if 1 represents winning, -1 represents losing, and 0 represents a tie, the *payoff matrix* for Player A is the following.

		Player B		
		Rock	Paper	Scissors
Player A	Rock	0	-1	1
	Paper	1	0	-1
	Scissors	-1	1	0

      iii. Similarly, the payoff matrix for Player B is the following.

		Player B		
		Rock	Paper	Scissors
Player A	Rock	0	1	-1
	Paper	-1	0	1
	Scissors	1	-1	0

      iv. This is called a *total conflict* game because the sum of the two matrices is a constant matrix.
         1. In other words, nobody can win without the other person losing.
         2. Also, the value of winning is all the same, and the value of losing is all the same.
      v. This is also called a *zero-sum* game because the sum of the two matrices is the zero matrix.
      vi. What strategy should a player choose?
         1. It's hard to say in this game.
         2. Probably a randomized strategy; otherwise, your opponent would be able to pick up on what you always choose.
   b. A second example
      i. Suppose two politicians are campaigning for political office. In the final two days of the campaign, they have the choice of spending time in two important cities in the state: Tacoma and Spokane. They could spend time in either or both cities. The scheduling for the second day must be set before knowledge of the opponent's first-day schedule becomes known.
      ii. The three strategies are the following.
         1. Strategy 1: spend one day in each city
         2. Strategy 2: spend both days in Tacoma
         3. Strategy 3: spend both days in Spokane
      iii. Suppose Politician A's estimated payoff table is the following, where the numbers are in units of 1000 votes.

		Player B		
		1	2	3
Player A	1	1	2	4
	2	1	0	5
	3	0	1	-1

iv. Politician B's estimated payoff table would be the negative of this.
v. What strategies should the players choose?
vi. Dominance
    1. A strategy A *dominates* a strategy B if every outcome in A is at least as good as every outcome in B.
    2. Game theory assumes that a rational player will never play a dominated strategy in a total conflict game.
    3. With Rock, Paper, Scissors, there are no dominated strategies.
vii. With these assumptions, we can see that Strategy 3 is dominated by Strategy 1 for Player A. This simplifies the payoff table for Player A like so, since Player 2 assumes that Player 1 is rational as well:

		Player B		
		1	2	3
Player A	1	1	2	4
	2	1	0	5

viii. For this payoff table, Strategy 3 is dominated by Strategy 1 for Player B. This simplifies the payoff table like so:

		Player B	
		1	2
Player A	1	1	2
	2	1	0

ix. Now, Strategy 2 is dominated by Strategy 1 for Player A. This simplifies the payoff table like so:

		Player B	
		1	2
Player A	1	1	2

x. Finally, Strategy 2 is dominated by Strategy 1 for Player B. Therefore, if both politician players are rational, they will both choose Strategy 1.
xi. The *value* of a game is the payoff to player A.
    1. The value of this game is 1.
    2. This is not a *fair* game because the value of the game is not zero.
    3. And we see that if both players play rationally, Politician A will pick up an extra 1000 votes in the last two days of the campaign.
xii. A *Nash equilibrium* is an outcome where neither player can benefit by departing unilaterally from its strategy associated with that outcome.
xiii. Players A and B both choosing Strategy 1 is a Nash equilibrium.

II. Partial conflict games
    a. An example of a partial conflict game is the classic "prisoner's dilemma."
        i. Suppose Katniss and Peeta are arrested. The Capitol has enough evidence to convict both for sabotage, but not enough to prosecute either for treason.
            1. If both Katniss and Peeta confess and betray each other, each serves twenty years in prison.
            2. If Katniss betrays Peeta and joins with the Capitol, but Peeta remains silent, Katniss will be pardoned and set free, and Peeta will be executed. The reverse holds as well.
            3. If both Katniss and Peeta remain silent, each will serve five years in prison for sabotage.

ii. The combined payoff table is the following, with Katniss's payoff as the first entry and Peeta's as the second.

		Peeta	
		Betray	Silent
Katniss	Betray	(20,20)	(0,100)
	Silent	(100,0)	(5,5)

iii. This game is a *partial conflict* game because the sum of the two payoff tables is not constant.

iv. Analysis
   1. The best option in terms of the sum of the punishments inflicted is for both to remain silent.
   2. However, Betrayal dominates Silence as a strategy for Katniss. The same is true for Peeta.
   3. Thus if both behave rationally, they betray each other.
   4. Betrayal is the only Nash equilibrium.

III. More variations on the political example

a. Suppose the payoff table for Player A in the political example looks like this instead:

		Player B		
		1	2	3
	1	-3	-2	6
Player A	2	2	0	2
	3	5	-2	-4

i. This game does not have dominated strategies, so what would a rational player do in this case?

ii. The usual assumption is that each player will know or eventually know their opponent's strategy.
   1. If both players are playing the game repeatedly, then it's likely that your opponent will figure out your strategy eventually.
   2. Or perhaps, even if only playing once, your opponent could guess or figure out your strategy in advance.
   3. If your opponent knows your strategy, then they will choose a strategy that minimizes your payoff.
   4. Given this, the usual assumption is that a rational player will choose the strategy that maximizes their minimum payoff.
   5. This is called the *maximin criterion.*
   6. Other comments
      - This is a fairly risk-averse assumption.
      - It's the strategy that provides the best guarantee of a payoff.
      - Some texts (including ours) call this the *minimax criterion* because the problem is framed as minimizing your maximum loss.

iii. Let's analyze Player A's options.
   1. Strategy 1's minimum payoff is -3, Strategy 2's is 0, and Strategy 3's is -4.
   2. Thus Player A will choose his Strategy 2.

iv. Similarly, Player B will choose the option that maximizes her minimum payoff.
   1. In terms of Player A's payoff table, this is equivalent to minimizing Player A's maximum payoff.
   2. Strategy 1's maximum payoff is 5, Strategy 2's is 0, and Strategy 3's is 6.

3. Thus Player B will choose her Strategy 2.

v. The resulting payoff to Player 1 will be 0.

vi. Notice that this point is a Nash equilibrium.

- If both players are playing this strategy, then neither will achieve a gain in payoff by switching strategies.
- Your text also calls this a *saddle point*, as it's a minimum value for Player A's row and a maximum value for Player B's column.

b. Now, suppose the payoff table for Player A looks like this:

		Player B		
		1	2	3
Player A	1	0	-2	2
	2	5	4	-3
	3	2	3	-4

i. Under the maximin criterion, Player A chooses his Strategy 1, and Player B chooses her Strategy 3.

ii. This results in a payoff of 2 to Player A.

iii. However, this does not represent a Nash equilibrium.

1. Player 2 would be able to switch to Strategy 2 for a better payoff of -2.
2. Player 1, knowing that Player 2 is rational, and anticipating this, could switch to Strategy 2 for a payoff of 4.
3. Player 2, anticipating this, could switch to Strategy 3 for a payoff of -3.
4. And Player 1, anticipating this, could switch back to Strategy 1 for a payoff of 2.
5. The logical cycle continues.

iv. The maximin criterion results in an *unstable* solution for this game.

v. In fact, this game has no Nash equilibrium.

vi. The best solution for the players here likely follows a mixed strategy (just as with Rock, Paper, Scissors).

IV. Mixed strategies

a. Let's allow the two players to have mixed strategies in the last example, and analyze the game again using the maximin criterion.

b. Strategy 3 for Player A is dominated by Strategy 2, so let's rule that out and obtain the following table:

		Player B		
		1	2	3
Player A	1	0	-2	2
	2	5	4	-3

c. Analysis for Player A

i. Let's suppose Player A uses Strategy 1 with probability x_1 and Strategy 2 with probability x_2.

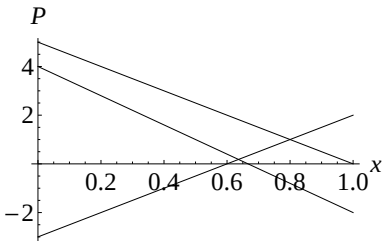
1. We have $x_1 + x_2 = 1$, $x_1, x_2 \geq 0$
2. Thus $x_2 = 1 - x_1$.

ii. The expected payoff for Player A depends on the strategies chosen by Player B.

1. If Player B chooses her Strategy 1, Player A's expected payoff is $0x_1 + 5(1 - x_1) = 5 - 5x_1$.
2. If Player B chooses her Strategy 2, Player A's expected payoff is $-2x_1 + 4(1 - x_1) = 4 - 6x_1$.

3. If Player B chooses her Strategy 3, Player A's expected payoff is $2x_1 - 3(1 - x_1) = -3 + 5x_1$.

iii. Graphically:



iv. Remember that Player B wants to choose a strategy that minimizes Player A's expected payoff.
   1. Under the maximin criterion, Player A will choose the strategy that maximizes his minimum expected value.
   2. This corresponds to an assumption that Player B knows (or can observe) Player A's strategy.
      - If Player B knows x_1, she will choose a pure strategy that corresponds to the bottom of the three lines for that value of x_1.
      - Player A can maximize his payoff under this idea by choosing the maximin value.

v. The logic here leads to the idea that Player A will choose the mixed strategy where $4 - 6x_1$ and $-3 + 5x_1$ intersect.
   1. This point is $x_1 = \dfrac{7}{11}$, $x_2 = \dfrac{4}{11}$.
   2. If Player A chooses this strategy, the payoff is $\dfrac{2}{11}$.

d. Now, Player B probably doesn't actually want to play either Strategy 2 or Strategy 3 deterministically.
   i. If Player A knows this, he can adjust his strategy accordingly.
   ii. Instead, Player B probably wants to play a mixed strategy (y_1, y_2, y_3).
   iii. What do the payoffs look like when we allow a mixed strategy for Player B?
      1. With a mixed strategy for Player B, the payoff for Player A is

         $$P = y_1(5 - 5x_1) + y_2(4 - 6x_1) + y_3(-3 + 5x_1),$$

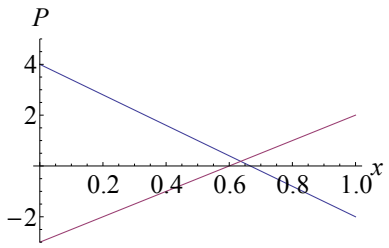
         where $y_1 + y_2 + y_3 = 1$, $y_1, y_2, y_3 \geq 0$.
      2. Note that this expression also accounts for the possibility of Player A playing the pure strategies $x_1 = 0$ and $x_1 = 1$.
      3. We know that Strategy 1 for Player B is never optimal, so we can assume $y_1 = 0$, simplifying this to

         $$P = y_2(4 - 6x_1) + y_3(-3 + 5x_1).$$

      4. Now, $P = 4 - 6x_1$ and $P = -3 + 5x_1$ are lines, and so $y_2(4 - 6x_1) + y_3(-3 + 5x_1)$ is the weighted average of two lines.
      5. What does the weighted average of two lines look like, graphically?
      6. For specific numbers y_2 and y_3, the weighted average will be

         $$P = 4y_2 - 3y_3 + (-6y_2 + 5y_3)x_1,$$

         which is itself a line.
      7. Graphically:



iv. Suppose that Player B chooses the mixed strategy $y_2 = y_3 = 1/2$.
   1. Then we have
   $$P = \frac{1}{2} - \frac{x_1}{2}.$$
   2. If Player A knows Player B's strategy, then Player A will choose the pure strategy $x_1 = 0$ and obtain a payoff of $1/2$.

v. Given this, how then can Player B choose a strategy that will minimize Player A's maximum payoff?
   1. Every choice of y_2 and y_3 results in a line that goes through the point where $x_1 = \frac{7}{11}$.
   2. So, for every choice of y_2 and y_3, the payoff will be at least $\frac{2}{11}$, the payoff for this mixed strategy of Player A.
   3. Any values of y_2 and y_3 that do not result in a constant expression for P (graphically, a horizontal line) will allow Player A to do better than a payoff of $\frac{2}{11}$.
   4. So Player B wants to find y_2 and y_3 that result in
   $$\frac{2}{11} = 4y_2 - 3y_3 + (-6y_2 + 5y_3)x_1$$
   for any choice of x_1.
   5. This means we can pick any two values of x_1 (since two points determine a line) and solve $y_2(4 - 6x_1) + y_3(-3 + 5x_1) = \frac{2}{11}$ to find y_2 and y_3.
   6. Taking $x_1 = 0$, $x_1 = 1$, we obtain
   $$4y_2 - 3y_3 = \frac{2}{11}$$
   $$-2y_2 + 2y_3 = \frac{2}{11}.$$
   7. Remember also that $y_3 = 1 - y_2$; this may simplify the calculations.
   8. The solution is $y_2 = \frac{5}{11}$ and $y_3 = \frac{6}{11}$, which (with $y_1 = 0$) represents Player B's optimal mixed strategy.
   9. Again, no matter what strategy (pure or mixed) Player A picks, this mixed strategy of Player B's assures that the payoff is no more than $2/11$.

vi. Generalizing, what we did here is to find values of y_1, y_2, y_3 such that
   $$\frac{2}{11} = P = y_1(5 - 5x_1) + y_2(4 - 6x_1) + y_3(-3 + 5x_1)$$
   regardless of the choice of x_1.

e. Check: Does this analysis still have Player A choosing the mixed strategy $\left(\frac{7}{11}, \frac{4}{11}\right)$?
   i. If Player A plays a different strategy, and Player B knows this, then Player B could pick a pure strategy with a lower payoff than 2/11, according to our earlier analysis.
   ii. So, yes, Player A will continue to play this strategy.
   iii. Thus the maximin criterion gives us a stable optimal solution (again, a *Nash equilibrium*): Neither player could change the payoff in their favor by unilaterally changing their strategy.

f. Maximin Theorem: If mixed strategies are allowed in a two-person total conflict game, the pair of mixed strategies that is optimal according to the maximin criterion is a Nash equilibrium.
   - The theorem is basically a consequence of duality theory for linear programs.

Homework: Chapter 10

- Section 10.1 #1, 2, 3abc. (Pair 1abc with 2abc, 1def with 2def, and 1ghi with 2ghi.)
- Section 10.2 #2g, 3b, 3e. (Use the solution approach we discussed in class.)
- Section 10.7 #3. (Hint: The entries in the payoff matrix are the expected payoffs for the various pairs of strategies. Also, there's a time component to this that you'll need to think about.)